A FINITE SET OF CODIMENSION-1-LIKE POINTS ON A SEPARATED SCHEME WITH COMPONENT-FINITE QC OPENS LANDS IN A COMMON AFFINE OPEN

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Allowed standard prerequisites (editorial).

- delivered companion: per-component reduction. delivered companion lemma-finite-set-codim-1-points-in-affine-per-component (S11-var-finite-set-per-component). Reduces the finitely-many-components case to the integral case.
- delivered companion: integral case (published). delivered companion lemma-finite-set-codim-1-points-in-affine (S11-var-finite-set-points-in-affine, tag 09NJ). Closes the integral case.

Source reference. Original lemma 09NM: sources/stacks-chow/tags.txt, source lines 12220–12220. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 09NJ: sources/stacks-chow/tags.txt, source lines 12217–12217. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Proposition 0.1. *Let X be a separated scheme such that every quasi-compact open has a finite number of irreducible components. Let $x_1, \ldots, x_r \in X$ be points such that $\mathcal{O}_{X,x_i}$ is Noetherian of dimension ≤ 1. Then there exists an affine open subscheme of X containing all of $x_1, \ldots, x_r$.*

Proof. We can replace X by a quasi-compact open containing $x_1, \ldots, x_r$ hence we may assume that X has finitely many irreducible components. By Lemma 09NM we reduce to the case where X is integral. This case is Lemma 09NJ. □

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